

Manisha Kaler

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EDUCATION

University of Massachusetts <i>Masters of Science in Computer Science</i> Coursework: Machine Learning, Performance Engineering, Data Science Fundamentals	Jan 2026 - May 2027 (Expected) 3.9/4.0
Indian Institute of Technology <i>Bachelors of Technology in Computer Science and Engineering</i> Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Networks, Operating Systems, Computer Organization and Architecture, Algorithms Design and Analysis, Theory of Computation, Computer Graphics	Aug 2017 - May 2021

WORK EXPERIENCE

Arun Data Processing <i>Software Engineer 2</i>	Feb 2023 - Dec 2025
- Built and shipped a Go-based PDF generation microservice for the Income Tax Department of India's production tax filing workflow, supporting 80M+ taxpayers. - Engineered schema-driven rendering logic that mapped nested JSON structures to reusable Go templates for tax forms, acknowledgements, and verification PDFs. - Benchmarked third-party HTML-to-PDF renderers and selected a compatible solution to improve PDF generation reliability for legacy RHEL Linux infrastructure. - Built a schema-driven code generation pipeline that generated Angular HTML, TypeScript, and service-layer code, reducing new workflow delivery time from 4 months to 3 weeks. - Developed 30+ Angular components and shared modules for validation, localization, and WCAG-aligned accessibility across online and offline filing flows. - Integrated 20+ client REST APIs end-to-end and packaged the Angular utility into cross-platform desktop executables using Wails.	

VISA

<i>Software Engineer</i>	Jul 2021 - Jan 2023
- Optimized ScalableRun, a global script execution platform managing 10,000+ servers, improving platform reliability through real-time validations and asynchronous processing. - Designed event-driven pub-sub systems using message queues to monitor Ansible Tower automation, surfacing execution metrics in Grafana dashboards to reduce MTTR by 30%. - Implemented the frontend for Ansible group management, enabling users to organize nested server groups for bulk script execution and improving deployment speed by 20%. - Built an IAM-based access validation layer to enforce user permissions across server groups, ensuring secure script executions across global Cloud PaaS infrastructure. - Built unit and integration tests for Spring Boot services using JUnit, Mockito, and Spring test utilities, achieving 95% code coverage.	

VISA

<i>Software Engineer Intern</i>	May 2020 - Jul 2020
- Developed a full-stack web application using Java and Visa Developer APIs to provide geo-targeted offer recommendations for small merchants. - Prototyped a functional payroll API solution in Java, digitizing merchant payment workflows for a global case challenge.	

PROJECT EXPERIENCE

Lecture Notes AI Assistant  <i>LLM-based Study Tool</i>	UMass Amherst
- Built an LLM-powered study assistant using Streamlit and Gemini 2.5 Flash-Lite to summarize transcripts, answer grounded questions, and generate study notes. - Implemented a structured storage layer with SQLAlchemy and SQLite to persist transcripts and support retrieval grounded responses.	
Cache-Optimized AVX Matrix Multiplication Kernel  <i>C++, AVX, OpenMP, Linux perf</i>	UMass Amherst
- Optimized C++ matrix multiplication with loop reordering, cache blocking, AVX vectorization, register-aware kernels, and OpenMP; achieved 4.48x single-thread and 2.44x 8-thread speedup on large workloads. - Used Linux perf and assembly inspection to analyze performance, reducing L1 cache miss rate from 72.47% to 6.56%, then to 1.25% with a cache-aware tiled kernel.	

TECHNICAL SKILLS

Languages: Go, Java, TypeScript, JavaScript, C/C++, SQL, HTML/CSS, Python
Frontend: Angular, React, Next.js, Bootstrap, Wails, Responsive Design
Backend & APIs: Spring Boot, Gin, RESTful APIs, Microservices
AI & LLM: Prompt Engineering, RAG Pipelines, Gemini API, Grounding
Cloud & DevOps: AWS (EC2, RDS, S3, Lambda), Docker, Jenkins, Ansible Tower, Linux, Git
Data & Tools: PostgreSQL, MySQL, MongoDB, Hazelcast, SQLite, SQLAlchemy, Grafana, perf
Testing & Process: Unit Testing, Integration Testing, Agile/Scrum